THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET B

Chapter 6: **Measuring Space: Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Taking $\pi = \frac{22}{7}$, the circumference of a circle of radius 14 cm is: 1

(a) 176 cm **(b)** 88 cm **(c)** 616 cm **(d)** 44 cm

2. The perimeter of a circle is 44 cm. Taking $\pi = \frac{22}{7}$, its radius is: 1

(a) 3.5 cm **(b)** 14 cm **(c)** 7 cm **(d)** 8 cm

3. The length of an arc subtending 60° at the centre of a circle of radius 21 cm ($\pi = \frac{22}{7}$) is: 1

(a) 44 cm **(b)** 11 cm **(c)** 22 cm **(d)** 132 cm

4. The length of an arc subtending 45° at the centre of a circle of radius 14 cm ($\pi = \frac{22}{7}$) is: 1

(a) 88 cm **(b)** 5.5 cm **(c)** 22 cm **(d)** 11 cm

5. By Heron's formula, the area of a triangle with sides 13 cm, 14 cm and 15 cm is: 1

(a) 90 cm^2 **(b)** 182 cm^2 **(c)** 84 cm^2 **(d)** 78 cm^2

6. The area of a parallelogram with base 12 cm and height 7 cm is: 1

(a) 42 cm^2 **(b)** 38 cm^2 **(c)** 96 cm^2 **(d)** 84 cm^2

7. The area of a triangle with base 16 cm and height 9 cm is: 1

(a) 72 cm^2 **(b)** 88 cm^2 **(c)** 144 cm^2 **(d)** 25 cm^2

8. By Brahmagupta's formula, the area of a cyclic quadrilateral with sides $5, 5, 12, 12$ is: 1

(a) 64 sq. units **(b)** 60 sq. units **(c)** 58 sq. units **(d)** 120 sq. units

9. Taking $\pi = \frac{22}{7}$, the area of a circle of radius 3.5 cm is: 1

(a) 22 cm^2 **(b)** 38.5 cm^2 **(c)** 77 cm^2 **(d)** 19.25 cm^2

10. The area of an equilateral triangle of side a is: 1

(a) $\frac{\sqrt{3}}{2}a^2$ **(b)** $\sqrt{3}a^2$ **(c)** $\frac{a^2}{2}$ **(d)** $\frac{\sqrt{3}}{4}a^2$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The diameter of a car tyre is 56 cm. How far does the car travel in 100 complete revolutions of the tyre? ($\pi = \frac{22}{7}$) 2

12. The perimeters of two circles are in the ratio 5 : 4. What is the ratio of their radii? 2

13. Find the perimeter of a semicircular plate of radius 7 cm (including the diameter). ($\pi = \frac{22}{7}$) 2

14. Find the area of an equilateral triangle of side 8 cm using Heron's formula. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the length of the altitude on the shortest side of a triangle with sides 7 cm, 15 cm and 20 cm. **3**

16. Prove that a median of a triangle divides it into two triangles of equal area. **3**

17. Use Heron's formula to show that the area of an equilateral triangle of side a is $\frac{\sqrt{3}}{4}a^2$. **3**

OR

Verify Brahmagupta's formula for a square of side a. **3**

18. Find the area of the cyclic quadrilateral with sides 25 cm, 39 cm, 52 cm and 60 cm. **3**

OR

Find the area of a circle of radius 21 cm and of the sector of angle 60° ($\pi = \frac{22}{7}$). **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Verify Brahmagupta's formula for a rectangle with sides a, b. **5**
(ii) Verify it for an isosceles trapezium with parallel sides 6 cm and 14 cm and legs 5 cm.

OR

Case study: A circular garden of radius 14 m has a 7 m wide path around it. ($\pi = \frac{22}{7}$) **5**
(i) Find the area of the garden.
(ii) Find the outer radius and the area of the garden together with the path.
(iii) Find the area of the path.
(iv) Find the cost of paving it at ₹50 per m^2.

20. **Case study:** A car tyre has diameter 56 cm. ($\pi = \frac{22}{7}$) **5**
(i) How far does the car move in one revolution?
(ii) How many revolutions does the tyre make in 1.76 km?
(iii) If the tyre turns 500 times per minute, find the speed in km/h.
(iv) A tyre of diameter 70 cm turns the same number of times; compare the distances covered.

OR

(i) Use Heron's formula to find the area of the triangle with sides 17 cm, 25 cm and 26 cm. **5**
(ii) Find the lengths of its three altitudes.
(iii) Which altitude is the longest?

END OF QUESTION PAPER — Total: 40 Marks

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